

INTERNATIONAL CONFERENCE ON RECENT TRENDS IN ADVANCED COMPUTING
2019, ICRTAC 2019**24X7 Lifeline Chip for Soldiers**Santosh Anand^{*1}, Somnath Sinha²Amrita School of Arts and Sciences, Mysuru
Amrita Vishwa Vidyapeetham, India**Abstract**

Soldiers spend a tough and risky life in battlefield. During an emergency situation they fight without any backup for saving their life. This causes the country to lose huge number of soldiers due to lack of technology rather than enemy attack. Location tracing of a soldier during such panic situation is required to get information of soldier's current situation. High end technological devices have been produced to help them manage various situations during their mission. In the current study one lightweight wearable device is developed that embed existing small devices onto a Chip (SoC) which piecewise integrate these devices to develop a user-friendly watch or batch or helmet. The proposed system enables a communication with the army base station to track the location and health status of soldiers. This is done through embedding GPS (Global positioning system) module with low power wireless body sensors in the system. Communication is set up two way: using antenna and using a GSM module. This improves reliability of the communication and makes it more fault tolerant. Hence the risk of non-functioning of the system in case of a failure in one part is reduced. Another important feature of this device is that it supports SDR (Software Define Radio) radio communication system used in Indian army.

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Keywords: Heartbeat sensor, Temperature sensor, GSM, GPS, Microcontroller, SDR.**1. Introduction**

"A soldier lives by chance, loves by choice and kills by profession". Soldiers are role model of any nation. Since so many years India is investing and invested heavily to get the high quality equipments for our Soldier. But technologies cannot control the nature, in some places temperatures are too much below than average which creates the snow slide. Due to this more soldiers have been killed and lost their life in the Siachen glacier region compared

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to enemy fire. Proposed system has ability to overcome from the drawbacks of military system. These devices will improve situational awareness who will exchange information using wireless networks. This system having two units: Soldier unit (Invented Chip) and Base unit. Soldier unit send the information and Base station (control room) will receive the information sent by soldier unit. Soldier unit consists of various sensors like heart beat sensor, temperature sensor (LM35), GSM module, GPS module, ATmega8 microcontroller. All the devices are embedded on microcontroller using embedded C language. Base unit collects all the information from the soldier unit and navigates them.

2. Literature Survey

In 2016, “Lance Naik Hanumantappa Koppad” was buried under the ice boulders up to 35 feet down due to adverse weather condition and his existence was known after five and half hours through radio set, but they were unable to track his location.

Fortunately, rescuers were able to access the telephone cable which could lead them to the location where the soldiers were buried. But it was not easy and couldn't help. So the team eventually kept digging with basic tools. Finally, after digging vigorously for six days, he was found. This incident inspired to work on the device which would overcome all the drawbacks of present military system. So identifying the location using sensor network [1][2019] in a indispensable requirements of Indian soldiers

Rubina. A.Shaikh, [2] [2012], in his work, investigated the care of critically ill patients. Considering in India, everyday many people get affected by heart attack, and many of them become more serious, because they did not get proper and timely help of doctors. This work is based on monitoring the health of remote patients, when they get discharged from hospital.

P.S. Kurhe, S.S. Agrawal, [3][2013], in their work designed a system that can track the soldier's location at any point of time. Additionally this system can communicate with the control room using GPS.

Pravin W, Sawan M, Ashu M, Arijit B, Anchal M, Ashish S,[4][2013], described an idea which can be used for tracking the position and the health status of soldier, which enables the army base station to plan the strategies according to current situation during war. In their system use of GPS tracking device and RF transceiver module provide the wireless system to monitor the health parameters and location tracking of soldiers. By using this system, the army base station will come to know the position of soldier and the health parameters such as body temperature and blood pressure of soldiers.

Rama SundariBattula, O.S.Khanna, [5] [2013], explained the Geographical routing protocol which is used for the communication between one node to another node to reach the destination. In this work each sensor node is location based.

Charles J, Charles C, Douglas H and Gary S, [6] [2017], in their work author descried about implementation and working of Blue Force Tracking system in United State (US).

Pavankumar, GhadgeVijay,PVidya,BSonali, [7][2014], described how to get the health condition and location of the soldier in their research work. Limitation of this work is that soldier can't communicate as well as soldier to soldier communications are not possible.

M.PranavSailesh, C.Vimal K, B.Cecil, B.M.Mangal Deep, P.Sivraj, [8] [2014], explained that how to monitor the health status of soldiers and how to reduce the casualties with the help of LapView program which will act as central commanding station. There are several limitation in this work such as size, power consumption as well as soldier to soldier communications.

Subhas C, [9] [2015], gave theidea of different kinds of light-weight, high performance WEARABLE sensors and devices that can be used for Health monitoring of Human Body in real time.

Deepti [10] [2015], described the applications, future trends and challenges of WSN, according to author WSN can be deployed in Military application for transmission of critical information of particular area in right time to right person. Limitation of this work is that the authors did not explain anything about implementation, what to do and where to implement.

Generally a soldier has to spend their own money to purchase GPS, Weapons light, temperature indicator etc. But US military system implemented GPS enable system BFT (Blue Force Tracking) which can provide the location of friendly or enemy forces. BFT can be also use to transfer the text and image messages. This system uses FBCB2 (Force XXI Battle Command Brigade and Below) Network to communicate between transmitter and receiver by continuously transmitting the locations. US military implemented BFT system in handheld device which reduces the size. This system can work only within the FBCB2 Network.

There are several problems which Indian Soldiers faces during wars and rescue operations, which will affect the war strategies, like:

1. There is lack of communication between the soldier and the control room.
2. Sometimes the soldiers are lost in the battlefield and are unable to track.
3. Various climatic conditions also affect the soldier's health and threaten their life.
4. Sometimes the soldiers are unable to convey their difficulties for immediate help during panic situations.

3. System on Chip

The proposed SoC having GPS module which can fetch the exact location of the soldiers who are lost and get injured in the battlefield and send to the Base Station (Control Room) by GSM or WSN. It can monitor the health status of the soldier by bio-medicals sensors (temperature and heartbeat sensor), which is embedded in this.

Proposed System Architecture

The below block diagram (Fig 1) shows the proposed chip (System on a chip). It represents the working of the Soldier Health Monitoring and Location Tracking System. It has two main parts, a soldier unit (SoC or Invented chip) and base unit (control room) which is shown in below Fig 2. This SoC consists of ATmega8 microcontroller, heart beat sensor, temperature sensor, GPS receiver, GSM transmitter and toggle switch.

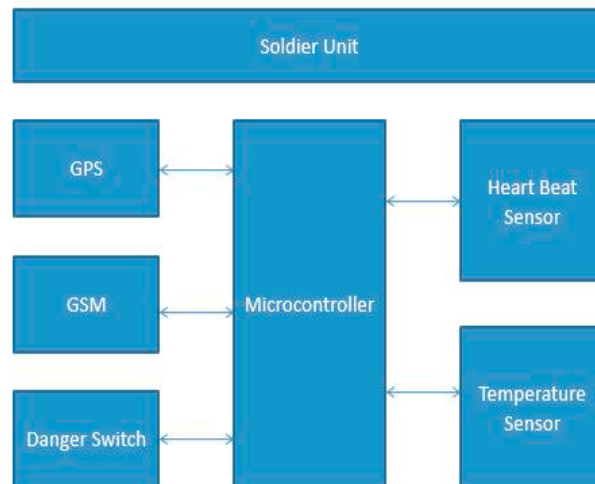


Fig 1: System unit

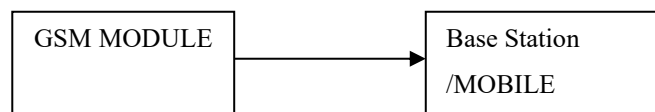


Fig 2: Base unit

4. Implementation

A. Interfacing GPS with Microcontroller

LCD (Liquid crystal display) is connected to the Microcontroller. The location of soldier will be displayed on LCD with latitude and longitude values. The receiver pin of GPS module is connected to max232 IC with ground. The GPS receiver will continuously transmit (latitude and longitude) the data using RS232 protocol. This LATITUDE and LONGITUDE values of location will be displayed on LCD. Range of GPS will be depend on the types of antenna, generally it is about 4-5 Km. GPS will send the signal to satellite and from satellite to Base station, these signals will travel at the speed of light. Below we represent (Fig 3) the GPS module which is used in this SoC.

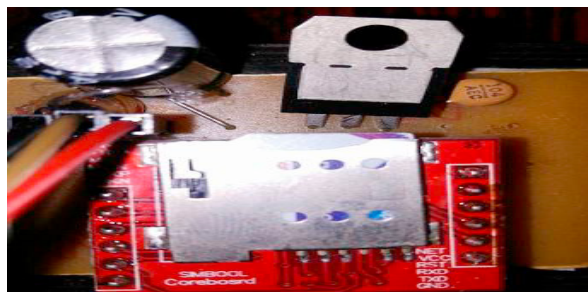


Fig 3: GPS Module

B. Interfacing with Heartbeat Sensor

In this SoC, optical fiber sensor is used to get the heartbeat of the soldier (Fig. 4). It is made to shine Infrared (IR) LED through the finger of the user and on other side will measure the slight changes in the light due to blood pump by a phototransistor to detect the pulse of the finger, a red LED flashes with each pulse. To make the sensor work properly one has to place the finger in between the emitter and detector, in the LED light side and phototransistor will be on the other side of the finger. Phototransistor used to obtain the flux emitted, when the blood pressure pulse by the finger then the resistance of the photo transistor will be slightly changed. In this work very high resistance resistor R1 is selected to achieve the accuracy.

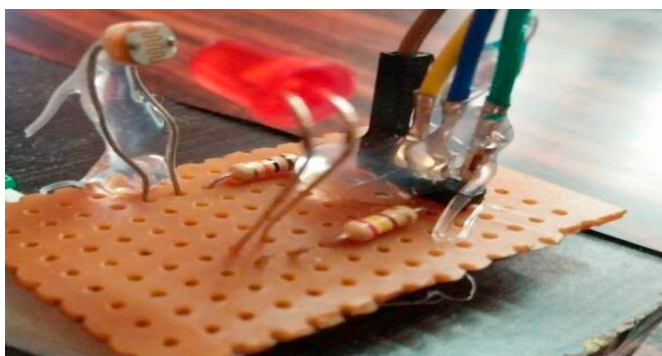


Fig 4: Heartbeat sensor

C. Interfacing with Temperature Sensor

In this SoC, analog temperature sensor (LM) is used to get the temperature of the environment (Fig. 5). This sensor can be easily use and implement with microcontroller (μC). It contains 3 pins which is used for input, output and ground. It uses solid state technique to measure the temperature. It amplifying the voltage change and generates the

analog signal which is directly proportional to the temperature. It will work from -40 degree C to 150 degree C. So it will be helpful for soldiers in glacier region as well as high temperature regions.



Fig 5: LM Temperature sensor

D. Software Define Radio

It is defined as the wireless communication device where the functionality of both receiver and transmitter can change or modified by software without any changes in Hardware. The major advantage of this system is that it can work on different frequencies not only on one frequency which is generated from hardware. It eliminates the use of resistors and capacitors because filtering algorithms are used to select the specified frequencies.

SDR provides algorithms for both downloadable and adaptable over the life span of the Radio Hardware. It makes more flexible and ideal for the Military communication. With SDR, the base station can monitor and communicate to the battlefield's soldiers over a large portion of spectrum of different protocols. SDR can also help to reduce the Chinese malware threats.

5. Communication Protocol

Two kinds of communication protocol is used in design of SoC to communicate from Soldier to control room that means soldier can either communicate through GSM or through antenna.

A. GSM Based Communication

In this protocol soldier node will directly communicate to control room by sending SMS.

i) Principle for GSM Interface

The GSM modem has the major roles in this circuit, because after collecting the data from sensor nodes it has to send to the base station. At present Indian Army is using Hardware based frequency generator for the communication, but now Indian Department of Defense are planning to use SDR for the communication to reduce the size, weight, easy to use, more secure and support for new generation technology. SDR is already successfully implemented by US Army.

There are following commands which can be used to implement the GSM with microcontroller (Fig. 6). These commands had to be coded with mobile number (Base station), than these commands will automatically executed according to the sensor nodes parameters and microcontroller instructions. In this design of SoC ATMEGA8 microcontroller is used. ATMEGA8 Microcontroller is the heart of this SoC, GSM will receive the commands from Microcontroller and perform the operations. These commands are the AT (attention) command, which is created by Microcontroller and send to the GSM modem. AT commands gives the instruction to the GSM modem to perform the following tasks.

1. Get the information of GSM modem such as manufacturer details (AT+CGMI), Software Version (AT+CGMR), Model number (AT+CGMM)
2. Information of subscriber.

3. Working status of GSM modem such as Signal strength (AT+CSQ) and battery level (AT+CBC).
4. Establish the data or voice communications.
5. Communication through message(SMS)
 - A. Send SMS:
 - Select SMS service: AT+CMSS
 - Send SMS: AT+CMGS
 - B. Read SMS :
 - List SMS: AT+CMGL
 - Read SMS: AT+CMGR
 - C. Write SMS: AT+CMGW
 - D. Delete SMS: AT+CMGW
 - E. Format SMS: AT+CMGF

Above commands having important roles while coding the SoC in embedded C. Currently I am using the Smart GSM modem which can support any kinds of SIM card provider for the Demo purpose but our SoC can support the GSM of Indian Army.

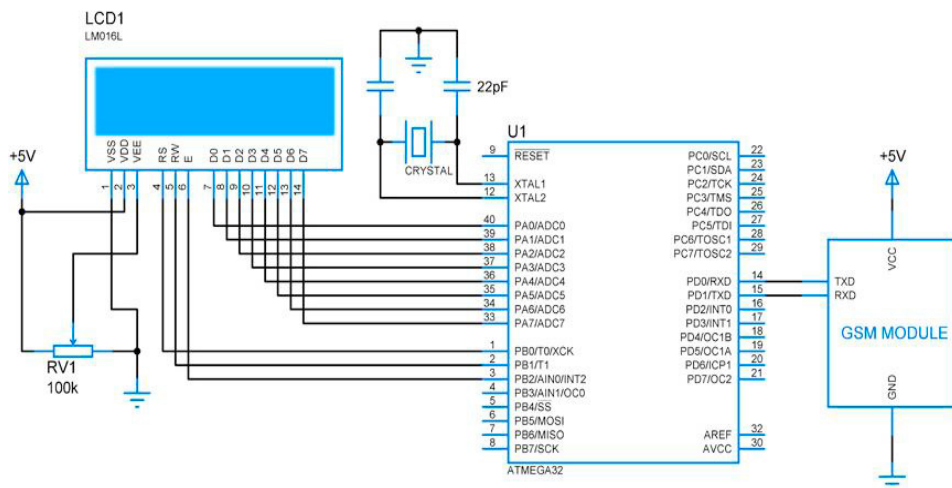


Fig 6: Interfacing GSM with Microcontroller

ii) Algorithm: For Soldier Unit (SoC)

1. Connect with Power supply.
2. Initialize serial communication.
3. Initialize the LCD.
4. Get the danger switch status if it's ON, Immediate sends one predefined SMS to Base.
5. Display the message on LCD to check the SoC status
6. Get location from GPS receiver and send to the LCD.
7. Display the GPS information on LCD.
8. Get soldier body temperature status from LM sensor.
9. Get soldier heart rate by optical from sensor.
10. Get the current status of GSM Modem.
11. Send all information together such as location, temperature status and heart rate to the LCD and base station using GSM communication.
12. Receive data from sim800.

B. Antenna based Communication by neighbour node to reach Control Room:

In the case of failure of GSM due to unavailability of network or some other problems then also soldier unit will be able to work and communicate to Control Room through the neighbour soldier units by antenna.

6. Result And Discussion

Outcomes of the work

- This system enables to monitor the health of soldiers using bio-medical sensors.
- It helps to minimize the time, search and rescue operation efforts of army control unit.
- Communication is enabled using GSM to the control unit.
- Soldier-to-soldier communication is also made possible using VBS(virtual backbone scheduling).
- It will support for upcoming communication technology SDR (Software Define Radio) of the Indian Army.
- Soldier can also see his own status on SoC, according to the information soldier can take precaution.
- Provides the dangerous switch to get the help in critical situations from control room (base station).

“24X7 Lifeline Chip for Soldiers” will produce the output to base station which is considered as Mobile phone. Figure 7 shows the final working system with two modules: Soldier Unit (SoC) and Base Unit (Mobile Phone). Mobile phone will receive the message when soldier will insert the finger in optical fiber sensor (heartbeat sensor), message will contain the location (LATITUDE and LONGITUDE), temperature and heartbeat of the soldier, below Figure8 shows the message which is sent by soldier to control room. There are two major advantage of this chip, first one is “When soldier will press the dangerous switch from the chip, message will send to the base station saying that soldier in dangerous zone” which is shown in below Figure 9. And second one is “Collocated communication can be made possible between soldiers.”

Generally soldier can only communicate to the control room but this project will help the soldier to communicate the nearest soldier so in the critical mission one soldier can help others immediately.



Fig 7: Final working system (SoC)

TEST CASE 1:

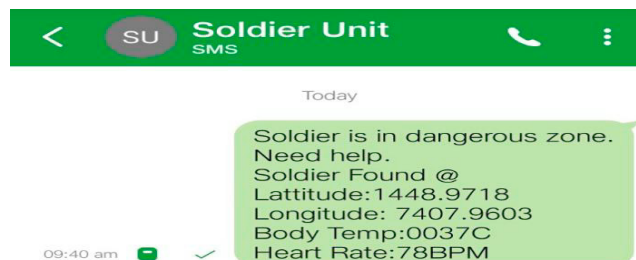


Fig 8: Soldier Units send message to Control Room

TEST CASE2:



Fig 9: Message from Soldier While Pressing Dangerous Key

7. APPLICATIONS

The system is low cost, easy to use, reliable and fast communication solution which provides a high level of safety and support for all kind to the soldiers in adverse environment. Continuous data logging can provide the analysis for different soldiers. The design also can fulfill all the requirements to support all the growing technologies used by Indian Army like SDR. The system once implemented will become helpful for Indian defence and also sometimes may helpful for remote health monitoring system.

8. CONCLUSION

This SoC will help the soldiers to communicate with base station and base station will get continuous information of soldiers. According to the information base station can instruct and plan for next move using wireless networks. The outcome of the integration of piecemeal components into a SoC develops lightweight package so it will consumes less energy. This system also solves the major challenges of power consumption in Wireless sensor networks. With proposed device, soldier can communicate to the control room any time. The proper navigation will too much helpful for the careful planning and co-ordination. Biosensors help us to notify status and active soldiers. It will also support to SDR for upcoming communication, which will be adaptable, clear, interoperable, fast, portable and effective communications. As future enhancement of this research oxygen sensor can be embedded to find out better location and for detection of explosive material another module can be added.

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